

Sprint 5 Documentation: “Final Polish”

AI-Driven Scaling of Strategic Technology Foresight

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Sprint 5: 22 June to 6 July 2026

Course: Digital Innovation Lab / Management & Digital Technologies II (LMU) | **Partner:** Rohde & Schwarz (PO: Dr. Andrew Schaefer)

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Abstract

Sprint 5 was the last content sprint before the closing presentations, and the aim was to take a working research prototype to a finished state. The team committed 55 story points across nine stories and completed 52; one story, the R&S local-inference integration, was blocked outside the team and carried over. The sprint also produced work that was never on the committed backlog, mainly the archetype and hybrid-structure analysis and the code that generates it, and that turned out to be the sprint’s main technical result. This document covers the product and sprint backlogs, the burndown, the sprint review, the story-point estimate against the outcome, the Scrum process and retrospective, the theoretical background, the results and their interpretation, the limitations and outlook, and the source code.

Contents

1	Product Backlog	2
2	Sprint Backlog	2
3	Burn-down Chart	2
4	Sprint Review	3
5	Story-Point Estimation vs. Reality	4
6	Scrum Process & Retrospective	5
7	Theoretical Foundation	6
8	Results	7
8.1	Scenario structure: before and after	7
8.2	Engine validation: the standard-matrix control	8
8.3	Interpretation & discussion: what it means for R&S	8
9	Limitations & Outlook	9
10	Source Code	9
	References	10

1 Product Backlog

The product is a domain-agnostic foresight engine: it ingests a knowledge base and derives a map of plausible futures for whatever domain that base describes. Regulatory spectrum monitoring, the Rohde & Schwarz field, is the running test case; the reusable engine itself is the deliverable. The backlog below is organised as nine epics along the pipeline, followed by a forward-looking backlog. Titles are reconstructed from the Jira project (DFF, board 34) and the codebase, so the exact wording may differ from the board.

Epic	Title	Status
DFF-2	Knowledge base & RAG ingestion (Chroma vector DB)	ongoing
DFF-3	BOM product decomposition (response drivers)	done
DFF-4	Trend scanning & driving-driver extraction (4 dimensions)	done
DFF-5	Driver merge & morphological (Zwicky) box	done
DFF-6	CIB cross-impact elicitation (persona panel + Delphi)	done
DFF-7	Scenario generation (fixed-point + combinatorial)	done
DFF-8	Structure & archetype analysis	done (this sprint)
DFF-9	Evaluation: MCDA (AHP/TOPSIS), temporal, strategic framing	done
DFF-10	Dashboard, UI & delivery	done (this sprint)
<i>Future backlog</i>		
DFF-23	Enrich the (thin) geopolitical corpus dimension	to do
DFF-24	MCDA score anchoring / de-compress evaluation scores	to do
DFF-25	Complete R&S local-inference endpoint integration (continues S7)	to do
DFF-26	Human-in-the-loop / stronger-model CIB validation	to do

2 Sprint Backlog

These are the nine stories committed for Sprint 5 (sprint board DFF, sprint id 69). The “Status @ 28 Jun” column is the mid-sprint checkpoint that anchors the burndown.

Key	Story	Assignee	SP	Status @ 28 Jun
DFF-11	S1 · Make pipeline domain-agnostic	Paul	8	Done
DFF-12	S2 · Fix positivity bias (split-scoring)	Christian	8	Done
DFF-13	S3 · Null-model significance test	Aditya	5	Done
DFF-17	S7 · Connect R&S local inference endpoints	Paul	3	To Do
DFF-18	S8 · ML technology-driver extraction	Aditya	8	In Progress
DFF-19	S9 · Polish UI	Adam	5	In Progress
DFF-20	S10 · LLM-as-a-Judge evaluation layer	Lovepreet	8	In Progress
DFF-21	S11 · Domain-agnostic on agriculture corpus	Christian	5	In Progress
DFF-22	S12 · Refine contrastive CCA	Paul	5	In Progress
Committed			55	
Done by 28 Jun			21	

3 Burn-down Chart

The actual line stays close to the ideal guideline through the sprint. Its last points are reconstructed from commit landing dates: the remaining stories were merged in the finalisation commit `bb90c09` on 8 July, two days past the official sprint end, which the retrospective (Section 6) picks up as a late integration. The team completed 52 of the 55 committed points; S7 (DFF-17, the R&S local-inference endpoints) was blocked externally and carried over.

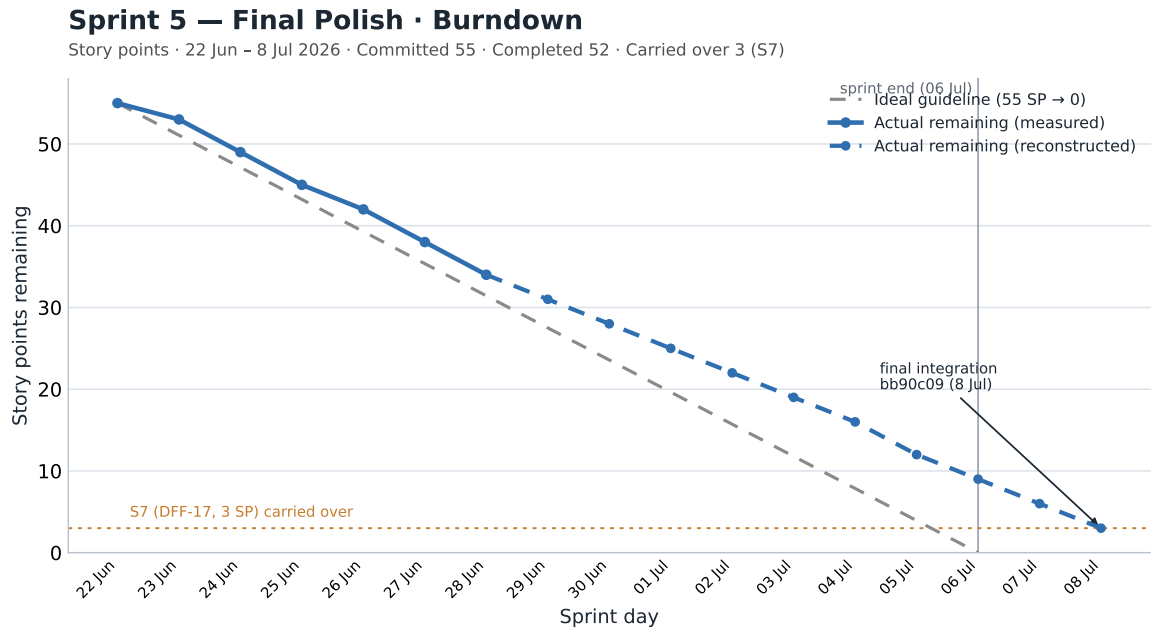


Figure 1: Sprint 5 burndown (story points). The actual line is measured for 22–28 June (Sprint5_Burndown.xlsx) and reconstructed from commit landing dates afterwards. It stays close to the ideal guideline and clears on 8 July with the finalisation commit `bb90c09`, two days after the official sprint end (6 July). It stops at 3 SP because story S7 (DFF-17) carried over. Of the 55 committed points, 52 were completed.

4 Sprint Review

Sprint goal

The goal was to bring the prototype to a finished product: repair the remaining quality problem in the cross-impact step, make the engine reusable across domains, and get the analysis behind a dashboard in time for the final presentations. The items below explain the reasoning behind each piece of work as well as what it does.

What was delivered

Ownership is noted per item; the committed backlog was spread across the whole team.

- **ML technology-driver extraction (S8, Aditya Purwar).** Aditya built an arXiv ingestion pipeline that pulls candidate technology drivers straight from the research literature with an ML/NLP stage. It complements the manual bottom-up BOM decomposition and grows the driver base on its own, so the driver set scales with the corpus instead of with analyst time. That is what the domain-agnostic goal needs: a new field should not require a fresh round of manual driver-hunting.
- **LLM-as-a-Judge evaluation layer (S10, Lovepreet Singh).** Lovepreet added an evidence-grounded auditor that scores each scenario and cites the source passages behind every score, then feeds those scores into the multi-criteria ranking (AHP weights, TOPSIS). R&S treats traceability (“Nachvollziehbarkeit”) as a hard requirement, so a score the model simply asserts is not enough; it has to point back to the corpus.
- **Cross-impact quality fix (S2, Christian Götsch).** Language models lean positive. Asked whether two technologies can coexist, they almost always say yes, so the earlier cross-impact matrix carried hardly any trade-offs and the scenario space collapsed onto a single consensus

future. Hand-tuning the prompt is brittle and tends to overshoot into fake pessimism, so Christian changed the aggregation instead: a strong minority trade-off is now kept in the panel result rather than smoothed out by the median. The share of negative couplings rose from 2% to 29%, inside the 20–30% band reported for real CIB matrices (Weimer-Jehle 2006).

- **Domain-agnostic engine (S1, S11; Paul Keck and Christian Götsch).** The demo topic was hard-coded in about fifteen places, so any other corpus produced nonsense. The Product Owner had made clear that the deliverable is a reusable framework, so the flow was turned around: the engine reads the corpus first, works out the domain profile itself, and passes that to every later step. A guard test fails the build if a domain term creeps back into the code. Docking an unrelated corpus on agriculture and field robotics produced a coherent analysis with no code changes.
- **Null-model significance test (S3, Aditya Purwar).** A scenario study only carries weight if the question “are there distinct scenario families?” gets an evidence-based answer. Aditya’s test compares the real field against a uniform-random field of the same shape and returns a significance (z) score, so the claim rests on a number instead of a hunch.
- **Dashboard and UI (S9, Adam Foster-Baird).** Adam’s React/FastAPI dashboard shows the landscape, the named archetypes, and the multi-method structure comparison next to each other, so a non-technical stakeholder can work through the result without opening a JSON file.

Main result

The main finding (Section 8) is that the scenario field is a hybrid. A dense core of a few well-separated scenario families sits inside a continuum, where the rest of the field shades gradually from one case to the next. Standard clustering misses the core; a density method on an ordinal encoding brings it out. Running the same method on a synthetic field that does contain real structure recovers clean clusters, which is why the “mostly-continuum” verdict on the real field reads as a feature of the domain: the tool finds structure when structure is there.

Review meeting with the Product Owner

The increments were shown to the Product Owner, Dr. Andrew Schaefer (R&S), at the sprint review. He accepted the domain-agnostic engine and the cross-domain demonstration on the agriculture corpus as the main increment of the sprint, and repeated his earlier steer: the reusable framework is what matters to him, and the spectrum results stay a test case. He again put weight on traceability, that every driver and scenario has to point back to a source. Two things followed for the backlog. First, keep hardening the framework’s generality and its evidence trail. Second, treat spectrum-specific polish as lower priority. The open R&S local-inference integration (S7) was noted as blocked on his side and moved to the next sprint.

5 Story-Point Estimation vs. Reality

Eight of the nine committed stories were finished. Only S7 carried over, blocked on an external R&S endpoint, and it has no commit because the dependency never arrived.

Key	Story	Committed	Completed	End status
DFF-11	S1 · Domain-agnostic pipeline	8	8	Done
DFF-12	S2 · Positivity-bias fix	8	8	Done
DFF-13	S3 · Null-model test	5	5	Done
DFF-17	S7 · R&S local inference endpoints	3	0	Carried over
DFF-18	S8 · ML technology-driver extraction	8	8	Done
DFF-19	S9 · UI polish	5	5	Done
DFF-20	S10 · LLM-as-a-Judge layer	8	8	Done
DFF-21	S11 · Agriculture-corpus proof	5	5	Done
DFF-22	S12 · Contrastive CCA refinement	5	5	Done
Total		55	52	3 carried over

Scope delivered beyond the commitment

The sprint also took on work that was never on the committed backlog, mostly the archetype and continuum analysis and the code needed to establish it. None of it was estimated in advance. A rough sizing after the fact comes to about **39 SP**:

Unplanned work delivered	≈SP	Landed in
Archetype module (HDBSCAN+ordinal, LLM-named clusters, continuum halo)	13	bb90c09
Multi-method structure comparison + structure referees	8	2d1583b
Temporal / DVI signal-maturity module	5	34e08ba
run_full.py one-command orchestrator	5	bb90c09
Corpus enrichment (arXiv + report PDFs) + engine-soundness validation	5	43969d2
LLM infra hardening (pooled model, timeouts, retry)	3	bb90c09
Total unplanned	≈39	

Estimation reflection. The committed 55 SP were mostly met, with 52 completed, but a similar amount of unplanned work was absorbed without ever being planned. That points to a planning miss: exploratory scope was under-counted and should have gone onto the board as estimated stories while it was happening. The retrospective (Section 6) lists the corrective actions.

6 Scrum Process & Retrospective

Roles

Following the course model, the team rotates the Scrum Master role each sprint. Paul Keck held it for Sprint 5 and owns this documentation. The Product Owner is external (Dr. Andrew Schaefer, R&S), and the other members work as a self-organising development team.

Release strategy

Work is integrated on the `feature/combinatorial-landscape` branch and checked by a continuous-integration workflow that runs `pytest` on every push. The definition of done for the sprint increment is a runnable pipeline plus dashboard: the whole analysis reproduces from one command (`run_full.py`), and the React front-end builds against the FastAPI back-end. Story points count as burned only once a story’s code has landed and its tests pass. Results are seeded so runs reproduce, and every claim traces back to source passages through propagated `source_chunk_ids`.

Cadence & attendance

Sprint 5 ran on two whole-team meetings, one for planning at the start and one for the review and retrospective at the end, plus a bi-daily stand-up every second day. Attendance at the stand-ups was uneven: three or fewer of the five members usually joined, and on two of them only the Scrum

Master showed up. The Jira board (DFF) acted as the asynchronous record so absent members could pick up status without a live meeting.

Work distribution

The committed backlog was owned across the team: Aditya Purwar (ML technology-driver extraction, null-model test), Lovepreet Singh (LLM-as-a-Judge evaluation layer), Christian Götsch (cross-impact de-biasing, agriculture-corpus proof), Adam Foster-Baird (dashboard and UI), and Paul Keck (domain-agnostic core, contrastive CCA). The extra, unplanned work of about 39 SP was carried mostly by the Scrum Master with AI assistance. Concentrating it on one person is a bus-factor risk, which the retrospective picks up below.

Retrospective

- **What went well.** The mid-sprint scope clarification with the PO saved weeks of effort on the wrong deliverable. The dissent-preserving CIB fix and the engine-soundness check settled two open worries with measured numbers. The one-command orchestrator made the pipeline easy to demonstrate end to end.
- **What to improve.** Stand-up attendance was thin, twice down to the Scrum Master alone, and coordination suffered for it. Too much of the unplanned scope landed on one person, which is a bus-factor risk for a team deliverable. Estimation under-counted the exploratory work, which produced a large overrun and a late integration crunch. And one committed story (S7) hung on an external party, so it should have been marked blocked at planning rather than committed.
- **Actions for the next sprint.** Protect a fixed stand-up slot and ask for a written async update from anyone who cannot attend. Pair on the analysis modules so knowledge is not siloed. Set aside explicit capacity for discovery and open stories for it as it appears. Keep externally blocked items off the commitment until the dependency is available.

7 Theoretical Foundation

The pipeline puts together established foresight and decision methods with current NLP, and each step rests on a method with a track record.

Scenario planning and foresight. The work sits in the tradition of exploratory scenario planning (Schwartz 1991; Kosow & Gaßner 2008; Amer et al. 2013), which lays out several internally consistent, plausible futures to support decisions under deep uncertainty.

Morphological analysis. The space of futures comes from a morphological box (the “Zwicky box”): a small set of drivers, each with a few ordered states, whose combinations define scenarios (Zwicky 1969; Ritchey 2011). The states are kept ordinal (optimistic → pessimistic) so that later analysis can use their order.

Cross-Impact Balance (CIB). Not every combination is coherent. Following Weimer-Jehle (2006), a CIB matrix records for each pair of driver states whether one promotes or inhibits the other, and consistent scenarios are the ones with low internal tension. We elicit the matrix from a panel of LLM personas over two Delphi-style rounds and keep dissent in the aggregation, since LLMs carry a documented positivity and sycophancy bias (Sharma et al. 2023) that would otherwise erase the trade-offs.

Retrieval-augmented generation. Every generative step draws on retrieved source passages (Lewis et al. 2020), which keeps the narratives tied to the corpus and makes them traceable.

Structure analysis. To judge whether distinct scenario families exist, we use the silhouette coefficient (Rousseeuw 1987) against a uniform-random null model, and we set a partitioning method (KMeans) beside a density method that can leave points unassigned (HDBSCAN; Campello et al. 2013). Layouts are drawn with UMAP (McInnes et al. 2018).

Multi-criteria evaluation. Scenarios are scored on several criteria and ranked with the Analytic Hierarchy Process for the criterion weights (Saaty 1980) and TOPSIS for the ranking (Hwang & Yoon 1981).

8 Results

8.1 Scenario structure: before and after

The main deliverable is a map of the scenario space together with an assessment of how much structure it actually has. Under the standard lens, one-hot encoding with KMeans, every point is forced into some cluster and the 120 combinatorial scenarios come out as a single cloud (silhouette about 0.07, which is the random-null floor). Switching to a density method on an ordinal encoding (HDBSCAN, which is allowed to leave points unassigned) pulls out a dense core of five named archetypes at a core silhouette of about 0.38, with roughly a third of the field left as a continuum halo.

Read the two silhouettes correctly. The 0.07 (KMeans over all points) and the 0.38 (HDBSCAN over the dense core only) are not the same measurement, so they should not be sold as a “0.07 to 0.38 improvement”. Put plainly: the standard lens hides a core that the right lens brings out, and about a third of the field really is a continuum.

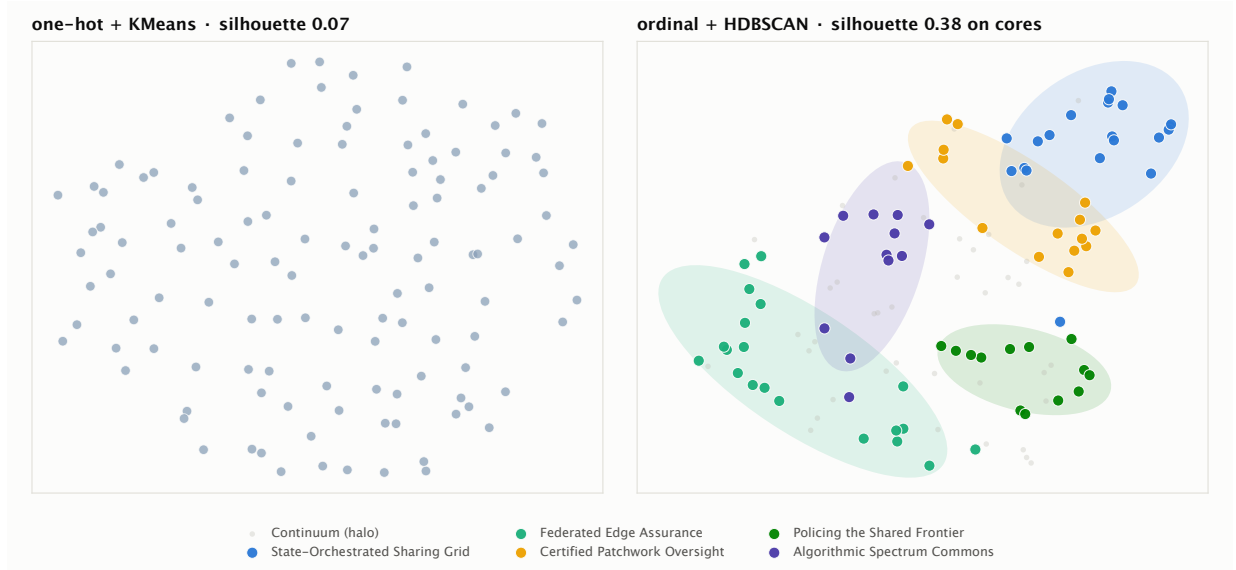


Figure 2: The scenario space under two lenses (the same 120 scenarios). Left: the standard view (one-hot with KMeans) forces every point into a cluster, so the field reads as a single cloud (silhouette 0.07). Right: a density method on the ordinal encoding (HDBSCAN) isolates five dense archetypes (core silhouette 0.38) and leaves about a third of the field as a grey continuum halo.

The five archetypes on the spectrum test case are **State-Orchestrated Sharing Grid** (20 scenarios), **Federated Edge Assurance** (20), **Certified Patchwork Oversight** (15), **Policing the Shared Frontier** (13), and **Algorithmic Spectrum Commons** (12); the remaining 40 make up the continuum.

8.2 Engine validation: the standard-matrix control

To check whether the “mostly-continuum” verdict says something about the domain or points to a limitation of the tool, the same clustering engine was run on a standard, ground-truth CIB matrix. This is a synthetic 8-driver matrix with two deliberately antagonistic blocks, where couplings promote within a block and inhibit across blocks. It stands in for the textbook gold-standard check in a form that is offline and deterministic. An uncoupled random matrix and the real spectrum field were run beside it:

Field	Silhouette	z -score	Eff. dim	Verdict
Standard matrix, coupled (positive control)	0.72	8.6	2.5	usable structure
Standard matrix, uncoupled (negative control)	0.22	−0.7	7.8	≈ uniform random
Real spectrum field (14 drivers, 120 scenarios)	0.07	1.4	30.0	≈ uniform random

What the control shows. On a matrix that really does contain structure the engine returns a clean silhouette of 0.72 ($z = 8.6$); on a random matrix it returns no usable structure ($z = -0.7$). The engine reads the input for what it is, so the low separability of the real field reads most naturally as a feature of the spectrum domain itself.

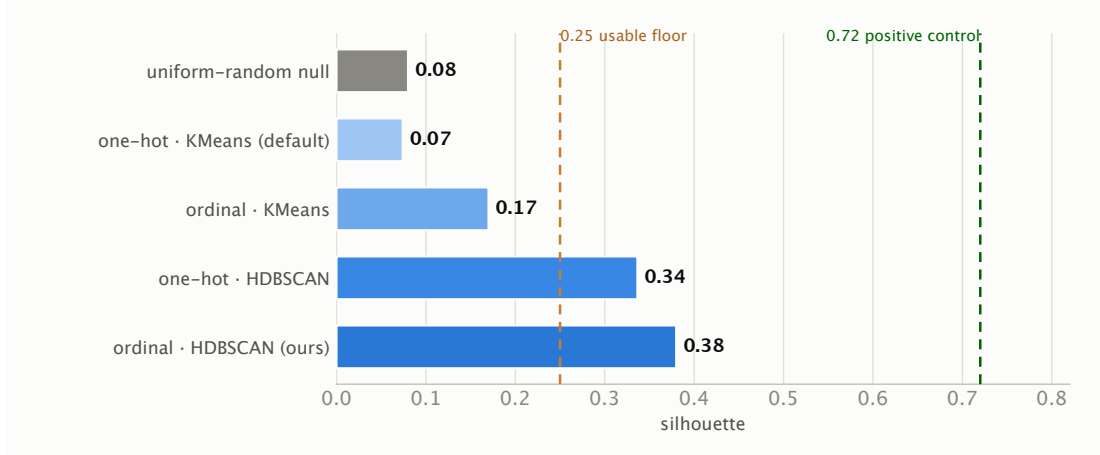


Figure 3: Silhouette of the same field under five lenses, with the 0.25 usable-cluster floor and the 0.72 synthetic positive control marked. The default lens (one-hot with KMeans) sits at the random-null floor; only the density method on the ordinal encoding clears the usable bar, and it is the same engine that reaches 0.72 on a field with real structure.

Two further measurements point the same way. A null-model test on the real combinatorial field gives $z = 2.62$ for the silhouette, which is above random noise but still below the 0.25 usable-cluster floor, exactly the hybrid picture. The CIB de-biasing moved the negative-coupling share from 2% under the positivity-biased setup to 29% under dissent-preserving aggregation, inside the literature’s 20–30% band. A contrastive elicitation prompt was also tried; it did not reliably raise the negative share, because the current drivers are largely complementary, and that negative result is reported here as well.

8.3 Interpretation & discussion: what it means for R&S

Each archetype describes a different world, and taken together they point to a few concrete implications:

- **State-Orchestrated Sharing Grid:** regulators run dynamic spectrum sharing from the centre. That plays to regulator-grade monitoring and enforcement, an R&S strength, though it also makes the government the dominant buyer.

- **Federated Edge Assurance:** distributed edge sensors with federated learning. Here R&S has to make its software-defined-radio and edge hardware interoperate with third-party sensor fleets, or it slides into the role of a component supplier.
- **Certified Patchwork Oversight:** fragmented, certification-heavy regional regimes, where demand moves toward compliance, calibration and certification tooling. That is a services opportunity.
- **Policing the Shared Frontier:** enforcement-centric, contested sharing. It sits closest to R&S’s current core of direction finding and illegal-transmitter detection and is the safest near-term bet.
- **Algorithmic Spectrum Commons:** AI-native, open spectrum, where hardware commoditises fastest and the sensible response is to move up the stack into AI-based signal classification and analytics.

The thread running through all five is a shift from hardware-defined toward software- and AI-defined monitoring. Since about a third of the field is a continuum, the safer strategy is to spread investment across the families and favour the software, AI and interoperability capabilities that pay off in more than one world; betting the whole roadmap on a single scenario would carry more risk.

9 Limitations & Outlook

- **Continuum halo.** The archetypes describe the dense core; roughly a third of scenarios stay in a continuum, and any archetype-based framing has to state that share.
- **Corpus skew.** The geopolitical driving dimension is still thin, with one driver; enriching it with more and opposing sources is scheduled as DFF-23.
- **MCDA compression.** The evaluation LLM tends to give similar impact and probability scores, so the ranking spreads less than it should; score anchoring is a separate lever (DFF-24).
- **Model variance.** Results are seeded but still depend on the LLM, so the figures are best quoted as ranges rather than single decimals.
- **Single-product BOM.** The bottom-up decomposition covers only the ESMW receiver so far; extending it to the wider R&S portfolio would widen the set of drivers.
- **External dependency (S7).** The R&S local-inference integration is blocked outside the team’s control and continues as DFF-25.

Outlook. The next steps are to finish the R&S endpoint integration, enrich the geopolitical corpus, add score anchoring to the evaluation, and bring in human-in-the-loop and stronger-model validation of the CIB matrix (DFF-26) so the cross-impact judgments are corroborated beyond the LLM panel.

10 Source Code

Repository. Smileodox/DILAB on GitHub, branch `feature/combinatorial-landscape`; Jira project DFF, board 34, sprint id 69 (private repo, access on request). The sprint deliverable is the finalisation commit `bb90c09`. The whole pipeline runs from a single command, and the test suite is green:

- Full run: `CIB_DISSENT_PRESERVING=1 uv run python run_full.py -skip-arxiv`

- Dashboard: `uv run uvicorn web.app:app -port 8000`
- Tests: `uv run pytest -q` (always `uv`, never `pip/venv`)

Key commits this sprint. Eight of the nine committed stories effectively landed in `bb90c09`; S7 has no commit, since it was blocked externally.

Hash	Date	Summary
b776715	27 Jun	Null-model structure test (S3)
48b3e61	27 Jun	Domain-agnostic pipeline + contrastive CCA (S1)
e850116	29 Jun	Multi-KB docking + interpretable landscape projection
031e490	05 Jul	<code>axis_role</code> layer-split (driving vs response)
2e9418e	05 Jul	Contrastive CIB mode (documented negative result)
2d1583b	05 Jul	Structure referees (collinearity + axis stability)
3ad97c6	05 Jul	Distribution-calibrated CCA filter (S12)
43969d2	05 Jul	Investigation scripts + capability proof
34e08ba	05 Jul	Temporal / DVI signal-maturity module
bb90c09	08 Jul	Finalize: archetypes, multi-method structure, UI, docs (50+ files)

Tooling and AI-assisted development. Toolchain: Python managed with `uv`; FastAPI and React for the dashboard; Azure OpenAI for the LLM and embeddings; Jira (company-managed project `DFF`) for the board and burndown. Development was AI-assisted (Claude Code) for implementation speed and refactoring, while the design decisions, validation and integration stayed with the team and were reviewed.

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